Notes: Athey & Ellison (QJE, 2011)
Position Auctions with Consumer Search

Contents

1 Big picture

- **Question.** How should sponsored-link position auctions be understood when the objects being auctioned are not generic slots, but advertisements that consumers search through rationally?
- **Core friction.** Consumers do not automatically click every ad. They infer advertiser quality from the order of sponsored links and click sequentially until either their need is met or expected quality falls below their search cost.
- **Main modelling move.** The paper embeds a consumer search model into a position auction model. Advertisers differ in their probability of satisfying a consumer's need, and the auction ranking becomes an information signal for consumers.
- **Auction equilibrium.** The generalized second-price style auction still has a monotone equilibrium: firms bid up to value to make the sponsored-link list, then shade bids when competing for higher ranks.
- **But values are endogenous.** A firm's value for a position depends on the qualities of all firms above it because those firms determine how many consumers reach lower links.
- **Welfare role of sponsored search.** Sponsored-link auctions create surplus not only by selling advertising slots, but by ordering firms in a way that makes consumer search more efficient.
- **Reserve prices.** Unlike standard auction models, reserve prices can increase consumer surplus and social surplus because they screen out low-quality links that would otherwise waste consumer search costs.
- **Click weighting.** Click-through weighting is not automatically efficient once consumer search and informational inference matter. It may select the wrong ads, reduce the informativeness of order, and induce obfuscatory ad text.
- **Policy takeaway.** Search-engine auction design affects consumers, advertisers, and platform profits through both allocation and information transmission. A rule that is revenue-improving or click-maximizing need not be welfare-improving.

2 Incremental contribution of the paper

- **Moves sponsored-search theory beyond slot auctions.** Earlier position-auction papers mostly treated slots as exogenous objects with click-through rates attached. Athey and Ellison instead model the fact that these slots contain advertisements that consumers interpret and search through.
- **Endogenizes click-through rates.** Clicks are not just fixed position effects. A link's click volume depends on the qualities of earlier links, the consumer's posterior beliefs, and the consumer's optimal stopping rule.
- **Adds a two-sided-market component to auction theory.** Auction rules affect advertiser bids, but they also affect consumer beliefs about ad quality. This feedback changes the welfare and revenue logic of standard auction design.
- **Reinterprets search engines as information intermediaries.** The paper emphasizes that sorting sponsored links by bid-relevant quality is valuable because it helps consumers search efficiently.
- **Revises standard reserve-price intuition.** In standard auctions, reserve prices raise seller revenue by excluding efficient trades. Here, reserves can raise welfare by preventing consumers from wasting costly clicks on low-quality links.
- **Qualifies the conventional wisdom on click-weighted auctions.** Weighting bids by click-through rates seems efficient when clicks are exogenous. The paper shows that with search costs, click weighting can distort information, reduce variety, and encourage misleading ad text.
- **Connects mechanism design to consumer inference.** The central incremental insight is that platform design is also information design: changing the auction changes what consumers infer from the displayed order.

3 Model primitives and variables

3.1 Consumers, needs, and search costs

- A continuum of consumers each has a need.
- A consumer receives utility 1 if the need is met and 0 otherwise.
- Consumers visit a search site displaying M sponsored links.
- Consumer j pays search/click cost s_j for each clicked link.
- Search costs are distributed according to G on $[0, 1]$ and are atomless in the baseline model.
- Consumers search optimally: they click only when the expected probability that the next advertiser meets their need is at least their search cost.

Interpretation: The consumer side is crucial because it makes ad positions valuable for endogenous reasons. A higher slot is valuable not only because it is seen earlier, but because consumers rationally infer that the firm in that slot is likely to be high quality.

3.2 Advertiser quality and payoffs

- There are N advertisers competing for $M < N$ sponsored positions.
- Advertiser i has quality q_i, the probability that the advertiser can meet a consumer's need.
- Qualities are independently drawn from a common distribution F on $[0, 1]$.
- Advertisers privately know their own quality.
- Advertiser payoff from successfully meeting a consumer's need is normalized to 1.
- Thus, the expected value of a click for advertiser i is q_i.

Interpretation: Quality is both a private value component for advertisers and an informational object for consumers. Consumers do not know qualities directly, so the position auction creates a public ranking signal.

3.3 Search engine and auction format

- The search engine sells M sponsored-link positions through an ascending position auction.
- The top M bidders are displayed in bid order.
- The kth highest bidder pays the $(k + 1)$st highest bid per click.
- The baseline model first studies the unweighted auction, then later considers click-weighted auctions similar to real-world sponsored-search systems.

Interpretation: The baseline auction is close to the generalized second-price auction, but the paper's environment differs sharply from standard GSP models because click-through rates are generated by consumer search rather than assumed exogenous.

3.4 Order statistics and consumer posterior beliefs

Let $q_{1:N}$ denote the highest quality draw, $q_{2:N}$ the second highest, and so on. Let z_k be a Bernoulli variable equal to one if the kth listed advertiser satisfies the consumer's need.

The expected quality of the kth link after failures above it is

$$\bar{q}_k \equiv E\left(q_{k:N} \mid z_1 = \cdots = z_{k-1} = 0\right). \tag{1}$$

Interpretation: A consumer who fails at the first link updates negatively about lower links because the event of failure is informative about the whole ordered list. The next click is valuable only if this posterior expectation remains high enough.

4 Theoretical specifications and main results

4.1 Optimal top-down search: Proposition 1

If firms are sorted by quality in equilibrium, consumers search from top to bottom and stop when the expected quality of the next site is below the search cost:

$$\text{consumer clicks link } k \quad \Longleftrightarrow \quad \bar{q}_k \geq s. \tag{2}$$

The conditional expectation is

$$\bar{q}_k = \frac{\int_0^1 x f^{k:N}(x) \Pr(z_1 = \cdots = z_{k-1} = 0 \mid q_{k:N} = x) dx}{\int_0^1 f^{k:N}(x) \Pr(z_1 = \cdots = z_{k-1} = 0 \mid q_{k:N} = x) dx}. \tag{3}$$

A firm in position k receives expected clicks proportional to

$$(1 - q_{1:N})(1 - q_{2:N}) \cdots (1 - q_{k-1:N}) G(\bar{q}_k). \tag{4}$$

Interpretation: Top-down search is not imposed mechanically; it follows from the sorting equilibrium. The failure of earlier links decreases the chance that lower links will work, so the consumer has a rational stopping rule.

4.2 Welfare from sorted versus unsorted lists: Proposition 2

If ads are randomly ordered and $\bar{q} = E(q_i)$, expected consumer surplus is

$$E(CS(s)) = \begin{cases} 0, & s \in [\bar{q}, 1], \\ (\bar{q} - s)\frac{1-(1-\bar{q})^M}{\bar{q}}, & s \in [0, \bar{q}]. \end{cases} \tag{5}$$

If ads are sorted by quality, consumers with search costs between the average quality and top-link quality click when they would not click in a random list.

Interpretation: Sorting creates welfare in two ways. Some consumers click because the top result is better than an average ad, and low-search-cost consumers find a match faster because high-quality firms are placed first.

4.3 Equilibrium bidding: Proposition 3

When more than M firms remain, firms bid up to quality to make the sponsored-link list. Once a firm is guaranteed a slot, it shades its bid. The symmetric equilibrium dropout point is

$$b^*(k, b^{k+1}; q) = \begin{cases} q, & k > M, \\ b^{k+1} + (q - b^{k+1}) \left[1 - \frac{G(\bar{q}_k)}{G(\bar{q}_{k-1})}(1-q)\right], & k \leq M. \end{cases} \tag{6}$$

Equivalently,

$$b^* = q - \frac{G(\bar{q}_k)}{G(\bar{q}_{k-1})}(1-q)(q - b^{k+1}). \tag{7}$$

Interpretation: The equilibrium resembles the standard sponsored-search result in broad form, but the values are no longer private and exogenous. The payoff to moving up depends on consumer search behavior, which depends on all higher advertisers' qualities.

4.4 Reserve prices and welfare

A reserve price r excludes advertisers whose quality or bid falls below the threshold. In a standard auction, reserves tend to trade off platform revenue against gains from trade. In this model, a reserve can improve welfare because low-quality ads generate wasteful consumer search costs.

For one-position lists with uniform quality and uniform search costs, the welfare-maximizing reserve price solves

$$r + r^2 + \cdots + r^N = \frac{N}{N+2}. \tag{8}$$

More generally, the paper shows that, with uniform search costs, consumer-surplus-maximizing and social-welfare-maximizing reserve policies coincide.

Interpretation: The reserve price is not only a rent-extraction tool. It changes consumers' beliefs about displayed ads. By committing not to show low-quality ads, the search engine can make consumers more willing to click.

4.5 Consumer-surplus and social-welfare alignment

For uniformly distributed search costs, the paper derives a strong alignment result: the policy maximizing consumer surplus also maximizes social welfare, and the sum of advertiser surplus and search engine profit is proportional to consumer surplus.

Interpretation: The result is useful because it turns a complicated three-party welfare problem into a simpler object. It also reveals the distributional conflict: once consumer surplus/social welfare is fixed, a profit-seeking search engine can raise its own surplus mainly by extracting advertiser surplus.

4.6 Click-weighted auctions

The paper then studies auctions in which bids are weighted by click-through rates. The standard intuition is that click weighting should improve efficiency because surplus is generated only when consumers click.

Athey and Ellison qualify this intuition:

- click-weighted auctions may choose ads that are too general rather than ads targeted to specific consumer needs;

- click-weighting can make the ordering less informative about advertiser quality;
- empirically estimated click-through weights may disadvantage high-quality firms because fewer consumers reach lower positions when earlier high-quality firms satisfy needs;
- click-weighted pricing can create incentives for firms to write broad or misleading ad text to attract extra clicks.

Interpretation: A high click-through rate is not the same as high social value. A vague ad can generate many clicks without satisfying many consumers. This is why click weighting can interact badly with consumer search costs.

4.7 Obfuscation incentive

When ad text can be informative, firms may choose how broad or precise their ad text is. In an unweighted per-click auction, firms dislike irrelevant clicks because they pay for them. In a click-weighted auction, steady-state payments can become insensitive to the firm's own click-through rate, creating incentives to increase clicks even when they waste consumer search time.

Interpretation: The paper's obfuscation result is an early theoretical warning that platform metrics can be gamed. If the platform rewards clicks without fully accounting for relevance, advertisers may design ads to attract low-value clicks.

4.8 Product variety

The base model assumes independent probabilities of meeting consumers' needs, but real keywords often cover heterogeneous needs. The paper extends the analysis to show that optimal weighting should give more weight to advertisers that increase product variety.

Interpretation: Diversity is a list-level property. The value of adding one ad depends on which other ads are displayed. Therefore, optimal ranking may require weights that depend on the whole set of sponsored links, not only on one advertiser's own characteristics.

5 Theoretical strategy and interpretation

5.1 What the theory identifies

The paper is theoretical rather than empirical. Its identification strategy is to isolate the mechanism by which auction rules shape consumer inference and search behavior. The key counterfactuals are:

- sorted lists versus random lists;
- no reserve versus positive reserve;
- unweighted auctions versus click-weighted auctions;
- informative ad text versus obfuscatory ad text;
- homogeneous needs versus heterogeneous needs requiring product variety.

Interpretation: Each counterfactual holds much of the environment fixed and changes the design rule, allowing the paper to map how design affects beliefs, search, bids, surplus, and distribution of rents.

5.2 Core assumptions

- Advertisers differ only in probability of meeting needs in the baseline.
- Consumers know the equilibrium sorting logic and search optimally.
- Advertisers know their own qualities.
- Search costs are heterogeneous across consumers.
- The search engine cannot directly observe all dimensions relevant to consumer welfare, so the ranking/auction design matters.

Interpretation: The assumptions are deliberately stylized to focus on information and search. The paper abstracts from advertiser pricing, production costs, organic search, and strategic product-market competition, but it captures the central fact that consumers interpret ad ordering.

5.3 Why the equilibrium is useful

The monotone bidding equilibrium creates a bridge between advertiser quality and consumer beliefs. Because higher-quality advertisers bid more, the list order communicates information. This makes sponsored-link auctions valuable as information mechanisms, not just allocation mechanisms.

Interpretation: Without monotone sorting, consumers could not infer quality from position. The paper therefore studies equilibria in which the auction performs the informational role that search engines are supposed to perform.

5.4 Threats and limits

- There may be other equilibria, including uninformative orderings.
- Consumers may not perform the exact Bayesian updating in the model.
- Real search engines also combine paid ads with organic results.
- Real advertisers differ in conversion value, prices, costs, and brand effects, not only in probability of meeting needs.
- The paper does not estimate the model using click data or advertiser bids.

Interpretation: These limits do not undercut the main contribution. They indicate that the model is best read as a mechanism analysis: it clarifies forces that standard exogenous-click models miss.

6 Figures

6.1 Figure A.1: Consumer Surplus with Sorted and Unsorted Links, $N = 4$

Read:

- The left panel shows consumer surplus under randomly ordered links.
- With random ordering, only consumers with search costs below the average quality click, and surplus declines roughly linearly.
- The right panel shows consumer surplus under quality-sorted links.

- Sorting raises surplus because high-search-cost consumers may still click the top high-quality link, and low-search-cost consumers find matches more quickly.
- The sorted curve lies above the unsorted curve for a wide range of search costs.

Interpretation: Figure A.1 visualizes the key informational role of sponsored search. The list is valuable because it tells consumers where to search first. The platform's auction is therefore performing a sorting function that creates consumer surplus.

Economic message: Sorting is not just a way to allocate advertiser exposure. It reduces consumers' search frictions and makes the advertising market socially valuable.

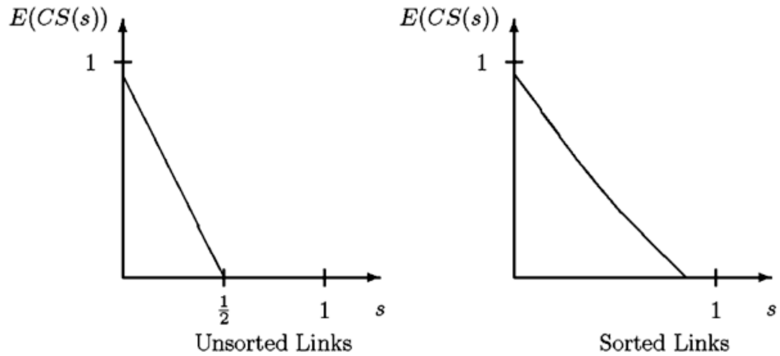

FIGURE A.1

Consumer Surplus with Sorted and Unsorted Links: $N = 4$

Figure A.1: Consumer Surplus with Sorted and Unsorted Links, $N = 4$

6.2 Figure A.2: Welfare and Distribution of Surplus for Two Specifications

Read:

- Figure A.2 plots consumer welfare, advertiser welfare, search-engine welfare, and a mixed objective against the reserve price.
- Vertical lines show the reserve price preferred by advertisers, consumers, the search engine, and the mixed objective.
- Advertisers generally prefer lower reserves because they dislike exclusion and surplus extraction.
- The search engine tends to prefer higher reserves because reserves raise revenue and improve consumers' willingness to search.
- Consumer surplus can be relatively flat over some reserve-price ranges, especially when reserves are rarely binding.
- The left and right panels illustrate that the trade-off depends on the distribution of advertiser qualities.

Interpretation: The figure shows that reserve-price policy is both an efficiency tool and a surplus-distribution tool. It can raise welfare by screening poor links, but it can also redistribute from advertisers to the search engine.

Economic message: Platform policy cannot be evaluated only by revenue or click volume. The reserve price changes who participates, what consumers infer, and how surplus is split between consumers, advertisers, and the platform.

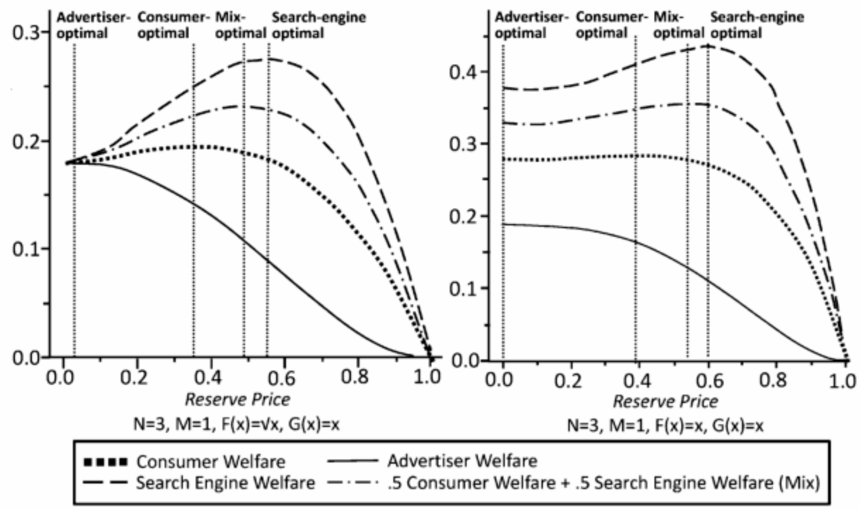

FIGURE A.2

Figure A.2: Welfare and Distribution of Surplus for Two Specifications

7 Interpretive synthesis

The paper's main synthesis is that sponsored-link auctions are information institutions. In a standard position-auction model, a slot has an exogenous click-through rate and an advertiser has an exogenous value per click. In Athey and Ellison's model, these objects are jointly determined by consumer inference. Consumers learn from order, order is generated by bidding, bidding depends on expected clicks, and expected clicks depend on consumer search.

This feedback explains why design tools can have surprising effects. A reserve price can raise welfare because it makes displayed ads more credible. Click weighting can reduce welfare because it may make the ranking less informative or reward broad ad text. Product variety matters because consumers have heterogeneous needs, so a list of sponsored links should not only contain high average quality but also serve multiple types of needs.

The paper also clarifies a conflict among market participants. Consumers and social welfare can be aligned under some conditions because consumers search optimally and bear search costs. The search engine's revenue objective can be close to consumer welfare in some ranges but can also redistribute surplus away from advertisers. Advertisers prefer rules that maximize expected profit, which may not maximize consumer surplus or social welfare.

8 Short summary

- The paper builds a model of sponsored-link auctions with rational consumer search.
- Advertisers differ in quality, interpreted as the probability of satisfying consumer needs.
- Consumers use the sponsored-link ordering as information about advertiser quality.

- In a monotone equilibrium, consumers search top down and stop when expected quality falls below search cost.
- Sorted sponsored links improve consumer welfare relative to random ordering.
- The auction equilibrium resembles standard GSP results, but values and click-through rates are endogenous.
- Reserve prices can raise consumer surplus and social welfare by screening out low-quality links.
- With uniform search costs, consumer-surplus and social-welfare maximizing policies coincide.
- Click-weighted auctions are not automatically efficient when consumers search and infer quality.
- Click-weighting can select overly general ads, reduce the informativeness of order, and encourage obfuscatory ad text.
- Product variety is a list-level design problem because the value of one ad depends on what other ads are shown.
- The main contribution is to treat sponsored-search auctions as mechanisms that allocate slots and transmit information.

9 Conclusion

9.1 Core conclusion

Athey and Ellison show that sponsored-link auctions cannot be evaluated as ordinary position auctions with fixed click-through rates. The displayed list affects consumer beliefs, consumer search, advertiser click values, platform revenue, and welfare. Auction design is therefore also information design.

9.2 Economic intuition

The intuition is simple: consumers click only if expected quality justifies the search cost. When the platform ranks ads by information correlated with quality, consumers search more efficiently. When the platform displays low-quality ads or uses a ranking rule that obscures quality, consumers waste search costs or stop searching too early. The auction rule matters because it determines what consumers infer from the order.

9.3 Incremental takeaway

The paper's key incremental contribution is to combine three elements that earlier sponsored-search work often separated: auction equilibrium, consumer search, and informational inference. This combination changes the conclusions about reserve prices, click-through weighting, ad text, and diversity of displayed links.

9.4 Policy and platform-design implication

A platform should not optimize only for clicks or short-run revenue. It should consider whether its ranking rule helps consumers identify relevant, high-quality advertisers. Rules that look efficient in exogenous-click models may be inefficient once clicks are generated by rational search and consumer beliefs.

9.5 Final takeaway

Sponsored search creates value by making search cheaper and more informative. The order of ads is a signal. Good auction design must preserve the credibility and usefulness of that signal.